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Physics-inspired priors for neutron star mergers improves equation of state constraints

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ABSTRACT

Gravitational-wave astronomy shows great promise in determining nuclear physics in a regime not accessible to terrestrial experiments. We introduce physics-inspired priors informed by nuclear theory and perturbative quantum chromodynamics calculations as well as astrophysical measurements of neutron star masses and radii. When these priors are used in gravitational-wave astrophysical inference, we show significant improvement on nuclear equation of state constraints. Applying these to the first two observed gravitational-wave binary neutron star mergers GW170817 and GW190425, constraints on the radius of a $1.4\,M_\odot$ neutron star improves from $R_{1.4} = 12.54^{+1.05}_{-1.54}$ km to $12.11^{+0.91}_{-1.11}$ km (90% confidence intervals). We also show these priors can be used to perform model selection between binary neutron star and neutron star-black hole mergers, although the results for GW190425 are inconclusive with a Bayes factor of 1.33 in favour of the binary neutron star merger hypothesis, representing only marginal evidence. We advocate for these physics-inspired priors to be used as standard in the literature, and provide open-source code for reproducibility and further adaptation of the method.

Keywords: neutron stars — gravitational waves — equation of state

1. INTRODUCTION

One of the main goals of gravitational wave (GW) astronomy is to infer constraints on the neutron star equation of state (EOS) from signals emitted by binary neutron star mergers. The baryon density in neutron stars and their mergers can reach several times the nuclear saturation density, $n_s = 0.16\,\mathrm{fm}^{-3}$, which is inaccessible to terrestrial experiments and ab initio calculations due to the infamous fermionic sign problem, preventing the direct solution of the fundamental theory of matter—Quantum Chromodynamics (QCD)—on the lattice. The major challenge lies in extracting information about the EOS at such densities from the GW signal of these events.

A common approach to astrophysical GW inference is to use agnostic priors for GW observables, assuming they are uncorrelated and uniformly distributed. For example, analyses of GW170817 (e.g., Abbott et al. 2017, 2018, 2019) and GW190425 (e.g., Abbott et al. 2020) have employed uniform priors on the chirp mass $\mathcal{M}$ and the dimensionless tidal deformability parameter $\tilde{\Lambda}$, the two most informative parameters in these events related to the neutron star EOS. However, the assumption of agnostic priors may be overly conservative, as basic physical principles such as causality of the EOS, along with consistency with nuclear physics and perturbative QCD imply a strong correlation between $\mathcal{M}$ and $\tilde{\Lambda}$ in neutron star binaries. Furthermore, imposing constraints from astrophysical measurements of neutron star masses and radii further tightens this correlation (Altiparmak et al. 2022a; Ecker & Rezzolla 2022).

Motivated by this, we introduce physics-inspired joint priors on $\mathcal{M}$ and $\tilde{\Lambda}$ that incorporate microphysical constraints from nuclear theory and perturbative QCD, as well as astrophysical constraints on neutron star masses and radii. We show an improvement on the measurement of the marginalized posterior on $\tilde{\Lambda}$ from the gravitational-wave observation of GW170817 from $\tilde{\Lambda} = 466^{+901}_{-158}$ with the agnostic prior to $\tilde{\Lambda} = 384^{+690}_{-226}$ with the new physics-inspired prior. This corresponds to an improvement in the neutron star radius measurement of an equivalent $1.4\,M_\odot$ star from $R_{1.4} = 12.54^{+1.05}_{-1.54}$ km to $12.11^{+0.91}_{-1.11}$ km For GW190425, the effective tidal deformabilty improves from $\tilde{\Lambda} = 381^{+1446}_{-110}$ to $\tilde{\Lambda} = 183^{+362}_{-82}$.

We also demonstrate the utility of these priors for distinguishing between the sources of Gravitational wave events. We perform model selection between a binary neutron star and neutron star – black hole origin of GW190425 and find a Bayes factor of 1.33 in favor of a binary neutron star origin.

In advocating for this prior becoming standard in the literature, we provide an open-source configuration for the BILBY Bayesian inference library (Ashton et al. 2019; Romero-Shaw et al. 2020) currently used for the majority of gravitational-wave parameter estimation by the LIGO-Virgo-KAGRA collaboration (Aasi et al. 2015; Acernese et al. 2015; Akutsu et al. 2020). This implements numerically the joint probability distribution as a constrained prior for $\mathcal{M}$ and $\tilde{\Lambda}$ as described in the next section.

2. METHODS

The construction of our physics-inspired prior closely follows that of Altiparmak et al. (2022b), which we briefly review here. We begin by building a large set of EOSs by stitching together various components. At the lowest densities ($n < 0.5\, n_s$) we adopt the Baym-Pethick-Sutherland (BPS) model (Baym et al. 1971) for the neutron-star crust. In the range $0.5\, n_s < n < 1.1\, n_s$, we randomly sample polytropes to span the range between the softest and stiffest EOSs from Hebeler et al. (2013). At high densities ($n \gtrsim 40\, n_s$), corresponding to a baryon chemical potential of $\mu = 2.6\,\mathrm{GeV}$, we impose the perturbative QCD constraint from Fraga et al. (2014) on the pressure $p(X, \mu)$ of cold quark matter, with the renormalization scale parameter X sampled uniformly in $[1, 4]$.

For the intermediate density range ($1.1\, n_s < n \lesssim 40\, n_s$), we use the parametrization method of Annala et al. (2020), which models the sound speed as a function of the chemical potential, $c_s^2(\mu)$, using piecewise-linear segments:

$$c_s^2(\mu) = \frac{(\mu_{i+1} - \mu)\, c_{s,i}^2 + (\mu - \mu_i)\, c_{s,i+1}^2}{\mu_{i+1} - \mu_i}\,, \qquad (1)$$

where μ_i and $c_{s,i}^2$ are parameters defining the i-th segment in the range $\mu_i \leq \mu \leq \mu_{i+1}$. Throughout this work, we adopt natural units where $c = G = 1$.

The number density follows as

$$n(\mu) = n_1 \exp\left(\int_{\mu_1}^{\mu} \frac{d\mu'}{\mu' c_s^2(\mu')} \right)\,, \qquad (2)$$

where $n_1 = 1.1\, n_s$, and $\mu_1 = \mu(n_1)$ is set by the corresponding polytrope EOS. The pressure is then obtained via

$$p(\mu) = p_1 + \int_{\mu_1}^{\mu} d\mu', n(\mu')\,, \qquad (3)$$

where the integration constant p_1 matches the pressure of the polytrope at $n = n_1$. We numerically integrate Eq. (3) using seven segments for $c_s^2(\mu)$.

Using this framework, we generate $\approx 10^6$ EOSs by randomly selecting the maximum sound speed $c_{s,\max}^2 \in [0, 1]$ and uniformly sampling the free parameters $\mu_i \in [\mu_1, \mu_{N+1}]$ (where $\mu_{N+1} = 2.6\,\mathrm{GeV}$) and $c_{s,i}^2 \in [0, c_{s,\max}^2]$. These EOSs are by construction consistent with nuclear theory and perturbative QCD uncertainties.

To incorporate astrophysical constraints, we solve the Tolman-Oppenheimer-Volkoff (TOV) equations for each EOS and retain only those satisfying the mass measurements of J0348+0432 (Antoniadis et al. 2013) ($M = 2.01 \pm 0.04\, M_\odot$) and J0740+6620 (Cromartie et al. 2020; Fonseca et al. 2021) ($M = 2.08 \pm 0.07\, M_\odot$), discarding EOSs with a maximum mass $M_{\mathrm{TOV}} < 2.0\, M_\odot$. In addition, we impose NICER radius constraints from J0740+6620 (Miller et al. 2021; Riley et al. 2021) and J0030+0451 (Riley et al. 2019; Miller et al. 2019), rejecting EOSs with $R < 10.75\,\mathrm{km}$ at $M = 2.0\, M_\odot$ or $R < 10.8\,\mathrm{km}$ at $M = 1.1\, M_\odot$. Unlike in Altiparmak et al. (2022a); Ecker & Rezzolla (2022) we do not impose any GW informed constraints on the binary tidal deformability in the construction of our proior.

In the neutron star binary case, the binary tidal deformability of the prior is computed as

$$\tilde{\Lambda}_{\mathrm{NSNS}} = \frac{16}{13} \frac{(12M_2 + M_1)M_1^4\Lambda_1 + (12M_1 + M_2)M_2^4\Lambda_2}{(M_1 + M_2)^5}\,, \qquad (4)$$

where M_i, R_i, and $\Lambda_i = \frac{2}{3} k_2 \left(R_i/M_i\right)^5$ $(i = 1, 2)$ denote the component masses, radii, and tidal deformabilities, with k_2 being the second tidal Love number. The chirp mass is given by $\mathcal{M}_{\mathrm{chirp}} = (M_1 M_2)^{3/5} (M_1 + M_2)^{-1/5}$ and we in the following we will use the mass ratio defined as $q = M_2/M_1$.

In the mixed binary case, i.e., where one binary component is a neutron star and the other one a black hole, the formula for the binary tidal deformability simplifies, because black holes have zero tidal deformability $\Lambda_{\mathrm{BH}} = 0$. The binary tidal deformability in this case becomes

$$\tilde{\Lambda}_{\mathrm{NSBH}} = \frac{16}{13} \frac{(12M_{\mathrm{BH}} + M_{\mathrm{NS}})M_{\mathrm{NS}}^4\Lambda_{\mathrm{NS}}}{(M_{\mathrm{NS}} + M_{\mathrm{BH}})^5}\,, \qquad (5)$$

where M_{NS} and Λ_{NS} are the neutron star mass and tidal defromability, respectively, and M_{BH} the black hole mass. The expressions for the chirp mass and mass ratio remain intact, with the understanding that $M_1 = M_{\mathrm{NS}}$ and $M_2 = M_{\mathrm{BH}}$.

2.1. *Binary neutron star mergers*

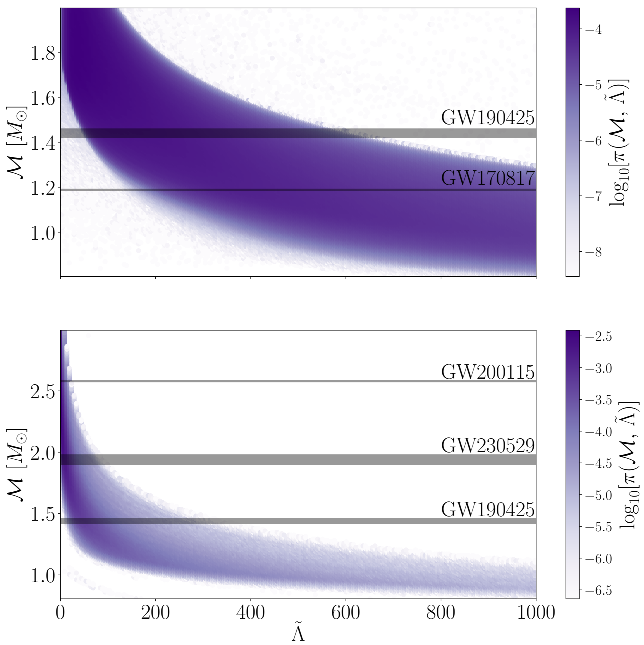

Figure 1. Coupled priors for the source-frame chirp mass $\mathcal{M}$ and dimensionless tidal deformability $\tilde{\Lambda}$. Top: prior for binary neutron star mergers. Bottom: prior for neutron star – black hole mergers. The horizontal grey bands are the 90% credible intervals for the measured chirp masses of GW170817 (Abbott et al. 2017), GW190425 (Abbott et al. 2020), GW200115 (Abbott et al. 2021) and GW230529 (Abac et al. 2024).

We use the probability distributions of the previous section as priors on chirp mass and effective tidal deformability as shown in Fig. 1. We implement this in Bilby using the `conditional_prior` function[1], interpolating the 200×200 grid to draw from a joint prior on $\mathcal{M}$ and $\tilde{\Lambda}$.

We sample directly in source-frame chirp mass, mass ratio $q = m_1/m_2$ and the two tidal deformability parameters $\tilde{\Lambda}$ and $\delta\Lambda$ (see Favata 2014; Wade et al. 2014, for explicit expressions for these quantities). We note that q is not a variable that depends on the equation of state, but instead priors *could* be chosen based on astrophysical formation scenarios. In this work, we take the standard approach and choose a prior on q that is uniform between $q = 0.125$ and 1 (e.g., Romero-Shaw et al. 2020). We further note that, while both $\tilde{\Lambda}$ and $\delta\tilde{\Lambda}$ are related to the equation of state, we only impose a prior here on $\tilde{\Lambda}$. In general, $\tilde{\Lambda}$ is measured much better than $\delta\tilde{\Lambda}$, which can be understood because $\tilde{\Lambda}$ enters waveform approximates at the 5^{th} post-Newtonian order, whereas $\delta\tilde{\Lambda}$ only enters at 6^{th} in linear combination with $\tilde{\Lambda}$ (Wade et al. 2014). In principle, one could set a combined prior on the three parameters ($\mathcal{M}$, $\tilde{\Lambda}$, $\delta\tilde{\Lambda}$), however, adding this latter parameter would not add significantly to parameter estimation in the relatively low signal-to-noise regime because of the above arguments.

2.2. *Neutron star – black hole mergers*

The parameterisation in the previous section does not go to high enough values of chirp mass because it is assumed both stars in the merger are neutron star. We therefore derive a new distribution for neutron star–black hole mergers by going back to the raw distributions of progenitor stellar masses and radii, and combining that progenitor with a black hole of mass m_1 and tidal deformability $\Lambda_1 = 0$. We use a uniform distributed mass ratio to generate our prior relationship. We have found more sophisticated mass-ratio distributions, such as those informed by state of the art binary population synthesis (e.g Broekgaarden et al. 2021) did not significantly impact the shape of the prior distribution. We demonstrate the independence of our results on the chosen population model in Appendix A. We choose once again to work with priors in $\mathcal{M} - \tilde{\Lambda}$, enforcing $\tilde{\Lambda}(\Lambda_2, \mathcal{M}, q)$ where Λ_2 is the tidal deformability of the lighter object in the binary, which we assume is the neutron star.

2.3. *Parameter estimation*

We perform Bayesian parameter estimation on the gravitational-wave strain data of the two (likely) binary neutron star merger events GW170817 and GW190425 using the strain data freely available from the GWOSC repository [2]. We utilize the Bilby Bayesian inference framework and priors on source–frame chirp mass and tidal–deformability implemented in Section 2. Priors for all renaming parameters assume the low-spin ($\chi \leq 0.05$) defaults as defined in Bilby (e.g Ashton et al. 2019, Romero-Shaw et al. 2020).

Parameter estimation for each run was performed using the dynesty nested sampler (Speagle 2020) and the IMRPhenomPv2_NRTidal waveform (Dietrich et al. 2019). Analysis of binary neutron star merger signals is significantly computationally expensive and as such we utilize pBilby (Smith et al. 2020), a parallel implementation of Bilby which uses the Message Passing Interface (MPI).

Since we are required to sample in the source–frame chirp mass instead of the observer frame, we must assume a cosmology. We use $H_0 = 67.66$ km s^{-1}, $\Omega_M =$

[1] git link to our code!?

[2] https://gwosc.org

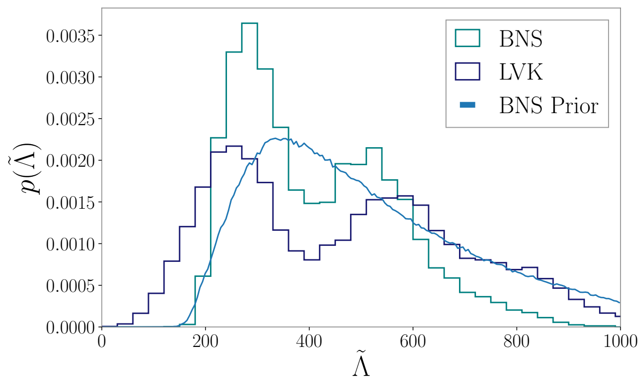

Figure 2. The posterior distributions of the tidal deformability $\tilde{\Lambda}$ obtained from Bayesian parameter estimation for GW170817. The posterior using a nuclear, microphysics, and astrophysics inspired prior is shown in teal, while the posterior using uniform priors from the LVK is shown in midnight blue. We show the (marginalized) prior with the light blue curve. Note, the posterior samples from the LVK show support up to $\tilde{\Lambda} = 2000$.

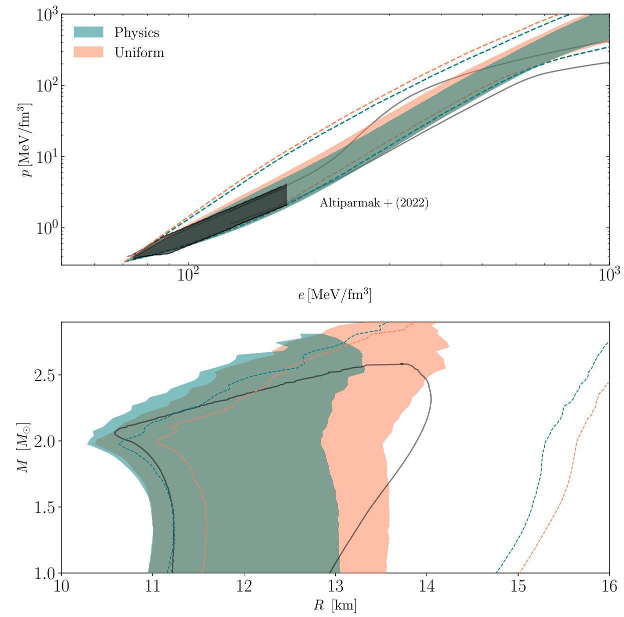

Figure 3. Posterior distirbutions of the cold nuclear equation of state from analysis of GW170817 using the spectral parameterisation (Lindblom 2010). The top panel shows the posterior distributions in pressure–density space, while the bottom panel shows the reconstructed mass–radius curves.The shaded regions show the 90% credible intervals of the posterior distributions obtained from the analysis using the physics inspired priors (teal) and uniform priors (orange). The dashed curves show the 90% credible intervals of the effective priors, whilst the grey curve shows the 90% credible interval of the true distribtuion used to generate the priors.

0.30966 informed by the Planck 2018 results (Planck Collaboration et al. 2020).

We employ both the binary neutron star and the neutron star – black hole merger priors in our analysis of GW190425. This facilitates a model selection analysis between a binary neutron star, and neutron star – black hole origin of GW190425. The details of this model selection analysis are described further in Appendix B. For GW170817 we perform parameter estimation using the binary neutron star prior only.

We reconstruct pressure–density and mass–radius curves from posterior samples of source–frame chirp mass, mass ratio, and effective tidal deformability by performing equation of state inference on the posterior samples of mass and tidal deformability. We utilise a four piece spectral pararmeterisation (Lindblom 2010), that is sitched to the SLy (Douchin & Haensel 2001) crust at low densities. Further details may be found in Appendix C.

3. RESULTS

Figure 2 shows the posterior distributions of intrinsic tidal parameter of GW170817 analyzed using the $\mathcal{M}-\tilde{\Lambda}$ feature (teal), when compared to the official posterior samples from the LVK (midnight blue). While we show only the tidal posteriors, we sample over all binary parameters. Posterior distributions for all other intrinsic and extrinsic binary parameters are identical.

Figure 3 shows the reconstructed pressure–density (top panel) and mass–radius (bottom panel) posteriors of GW170817. The shaded regions show the 90% credible intervals for the posteriors samples obtained using the physics inspired priors (teal) and uniform priors (orange). The dotted lines represent the 90% credible intervals of the effective priors on the physics prior and uniform prior when sampled with the spectral parameterisation. The gray curves show the 90% credible interval of the true equation of state distribution of our priors; i.e., the distribution used to generate our chirp mass–tidal deformability relationship. We find that the radius of a 1.4 $M_\odot$ neutron star can be constrained to $12.11^{+0.91}_{-1.11}$ km.

Figure 4 shows the posterior distribution of the intrinsic tidal parameter of GW190425 when analyzed using both the binary neutron star prior (teal) and neutron star – black hole prior (pink). We show the official posterior distribution of the LVK obtained using uniform priors in blue. Bayesian model selection between the binary neutron star and neutron star – black hole merger gives a Bayes factor of 1.33, marginally in favor of a binary neutron star merger origin.

Figure 5 shows the reconstructed pressure–density (top) and mass–radius (bottom) posteriors of

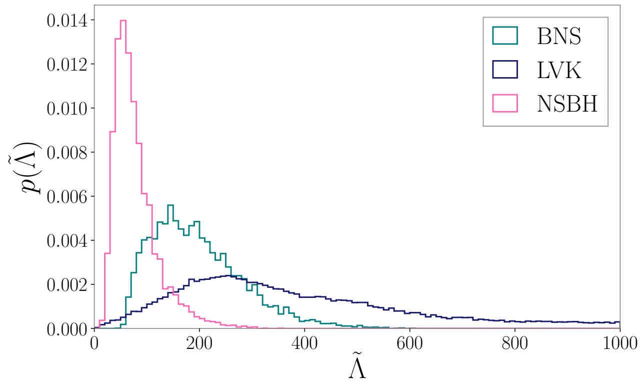

Figure 4. The posterior distributions of the tidal deformability $\tilde{\Lambda}$ obtained from Bayesian parameter estimation for G190425. The posterior using a nuclear, microphysics, and astrophysics inspired binary neutron star prior is shown in teal, and the corresponding neutron star – black hole prior is shown in pink. The posterior using uniform priors from the LVK is shown in midnight blue.

GW190425. Equation of state the 90% credible intervals of posteriors obtained using the binary neutron star physics inspired prior are shown in teal, while the neutron star–black hole posteriors are shown in pink. Once again the dashed curve represents our effective prior on the equation of state, while the gray curve shows the true prior. We find that the radius of 1.4 $M_\odot$ neutron star is constrained to $12.69^{+1.24}_{-1.22}$ km using the BNS prior, and $12.06^{+1.44}_{-1.15}$ km using the NSBH prior.

4. CONCLUSIONS

We have constructed new physics inspired priors for the analysis of gravitational-wave stran data of neutron star mergers. Analyis of the neutron star mergers GW170817 and GW190425 with our priors, constrains the effective tidal-deforambility of GW170817 to $\tilde{\Lambda} = 384^{+690}_{-226}$ and GW190425 to $\tilde{\Lambda} = 183^{+362}_{-82}$. Coresponding to a 37%, and 80% decrease in the width of 90% credible interval respectively when compared to uniform priors. Performing equation of state inference, we find that the radius of a $1.4 M_\odot$ neutron star can be constrained to $12.11^{+0.91}_{-1.11}$ km at a 90% credible interval. Which suggests a preference for softer/stiffer equations of state ...

Model selection between a binary neutron star and neutron star – black hole origin on GW190425, yields marginal support for a binary neutron star origin. However, such a result is not unexpected given the low signal to noise of the event and lack of significant tidal measurements.

In this work, we have constructed priors using binaries containing neutron stars with low spins ($\chi \leq 0.05$) only. However, analysis by the LVK typically involves both galactic "low-spin" and "high-spin" ($\chi \leq 0.89$) priors. We leave the development of "high-spin" binary neutron star and neutron star – black hole priors to future work.

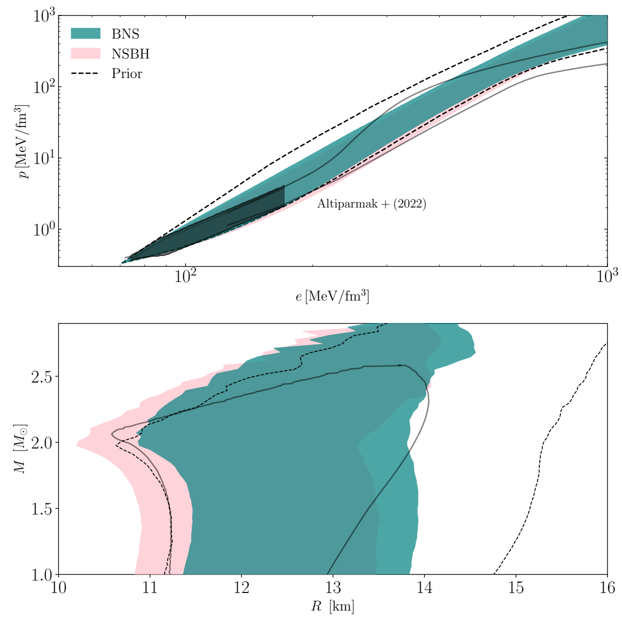

Figure 5. Posterior distirbutions of the cold nuclear equation of state from analysis of GW190425 using the spectral parameterisation (Lindblom 2010). The top panel shows the posterior distributions in pressure–density space, while the bottom panel shows the reconstructed mass–radius curves. The shaded regions show the 90% credible intervals of the posterior distributions obtained from the analysis using the physics inspired BNS priors (teal) and the NSBH prior (orange). The black dashed curve shows the 90% credible intervals of the effective priors, whilst the grey curve shows the 90% credible interval of the true distribtuion used to generate the priors.

APPENDIX

A. NEUTRON STAR – BLACK HOLE MERGER PRIOR GENERATION

We should detail the prior generation for neutron star – black hole here, in particular the independence of a mass ratio informed by population synthesis on the prior distributions and potentially the resultant posterior distributions.

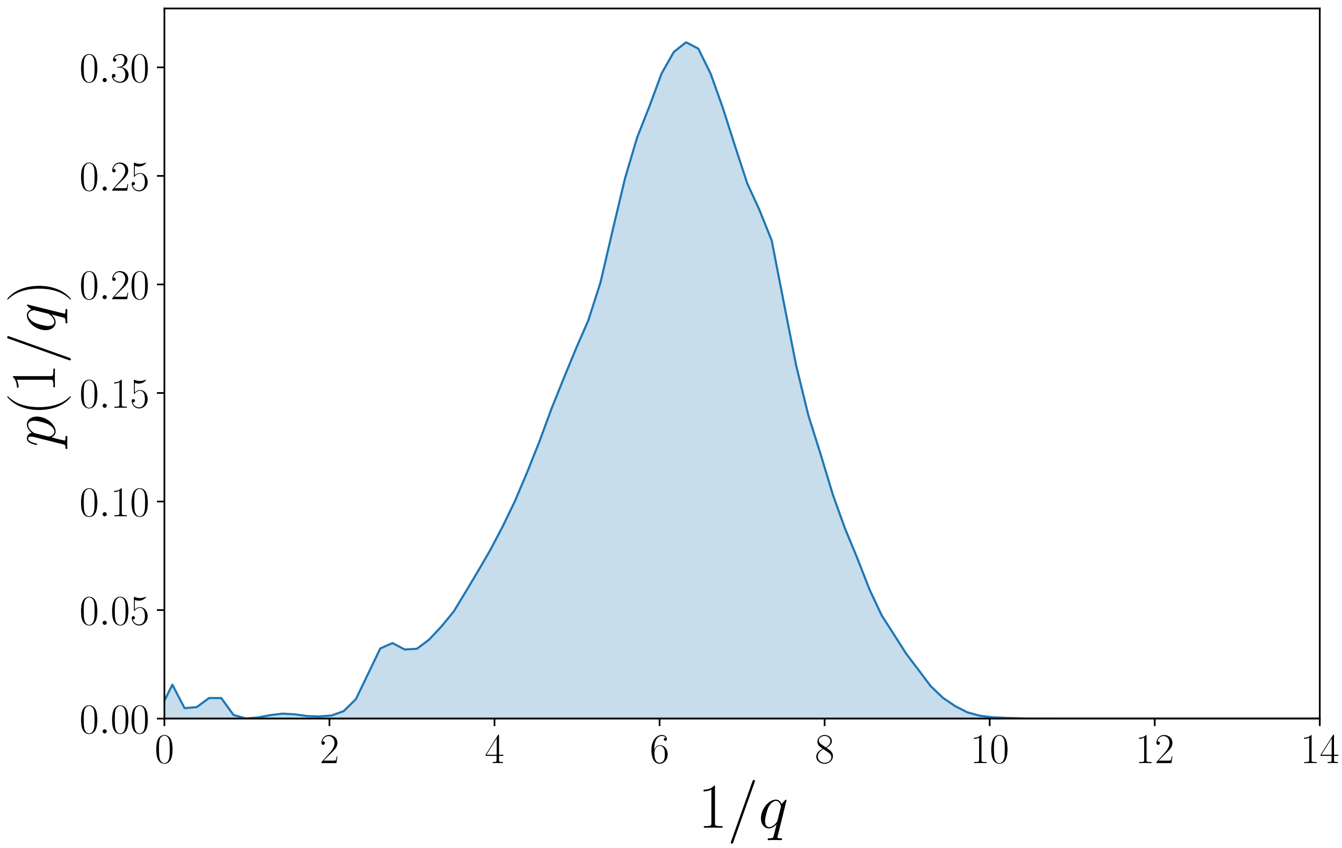

Figure 6. Probability density function of the mass ratio for detectable neutron star black hole mergers from (Broekgaarden et al. 2021).

Might be worth showing an extreme example to prove the point. PL: Probably delete this appendix; make a comment in the paper, and not include Fig. 6

We generate two relations, one with uniform priors on our population of BH - NS mergers, and one directly informed by state of the art binary population synthesis. We use the mass ratio distribution of Broekgaarden et al. (2021) (what particular model are we using?) for the generation of our population synthesis informed priors. Figure **??** shows the distribution in mass ratio used for the generation of the prior.

B. MODEL SELECTION

We perform parameter estimation of gravtitational-wave strain data using the DYNESTY nested sampler. Unlike standard MCMC samplers, DYNESTY calculates the evidence for each parameter estimation run. To perform Bayesian model selection between binary neutron star and neutron star– black hole mergers, we calculate the Bayes factor

$$\mathrm{BF}^{\mathrm{BNS}}_{\mathrm{NSBH}} = \frac{\mathcal{Z}_{\mathrm{BNS}}}{\mathcal{Z}_{\mathrm{NSBH}}}, \tag{B1}$$

where $\mathcal{Z}_{\mathrm{BNS}}$ and $\mathcal{Z}_{\mathrm{NSBH}}$ are the Bayesian evidences obtained for the parameter estimation runs using the binary neutron star and neutron star black hole priors respectively. Formally, for model selection we should be computing the odds ratio, which is defined as

$$\mathcal{O}^{\mathrm{BNS}}_{\mathrm{NSBH}} \equiv \frac{\mathcal{Z}_{\mathrm{BNS}}}{\mathcal{Z}_{\mathrm{NSBH}}} \frac{\pi_{\mathrm{BNS}}}{\pi_{\mathrm{NSBH}}}, \tag{B2}$$

where π_{BNS} and π_{NSBH} are the prior odds for each hypothesis; i.e our relative belief about which hypothesis is more likely. In this work we set the prior odds to unity, so the odds ratio is the Bayes factor. We could in principle construct an odds ratio where the prior odds was informed by the relative rates of binary neutron star and neutron – star black hole mergers. However since these rates are still highly uncertain, we choose the more agnostic approach of setting the prior odds to unity.

C. EQUATION OF STATE INFERENCE

The posterior on the equation of state from the gravitational-wave data is

$$p(\epsilon|d) = \frac{p(d|\epsilon)\pi(\epsilon)}{\mathcal{Z}}, \tag{C3}$$

where $\pi(\epsilon)$ is the prior on equation of state parameters ϵ, and $\mathcal{Z}$ is the evidence. The likelihood is given by

$$p(d|\epsilon) = \int d\theta\, p(d|\theta)\, \pi(\theta|\epsilon), \tag{C4}$$

where θ are our nuisance parameters which we marginalize over. Our likelihood reduces to

$$p(d|\epsilon) = \int \; d\theta \; p(d|\mathcal{M}, q, \tilde{\Lambda}) \; \pi(\mathcal{M}, q, \tilde{\Lambda}|\epsilon, \theta) \; d\mathcal{M} \; dq \; d\tilde{\Lambda}. \tag{C5}$$

Since we a dealing with a small number of events (i.e 1-2), we may ignore the modelling of overall neutron-star population parameters without introducing signficant biases into the equation of state inference (e.g Agathos et al. 2015; Wysocki et al. 2020; Landry et al. 2020). The marginalized likelihood transforms to

$$p(d|\epsilon) = \int \; p(d|\mathcal{M}, q, \tilde{\Lambda}) \; \pi(\mathcal{M}, q, \tilde{\Lambda}|\epsilon) \; d\mathcal{M} \; dq \; d\tilde{\Lambda}, \tag{C6}$$

We have two terms that need to be calculated; $\pi(\mathcal{M}, q, \tilde{\Lambda}|\epsilon)$ a prior on chirp mass, mass-ratio and $\tilde{\Lambda}$ for a given equation of state ϵ, and $p(d|\mathcal{M}, q, \tilde{\Lambda})$ the likelihood. Typically, this likelihood would be obtained by reweighting posterior samples of parameter estimation runs

$$p(d|\mathcal{M}, q, \tilde{\Lambda}) = \frac{p(\mathcal{M}, q, \tilde{\Lambda}|d)}{\pi_{\rm PE}(\mathcal{M}, q, \tilde{\Lambda})}, \tag{C7}$$

however if we did this we would be removing any of our prior information from the posterior samples. Since our prior on $\tilde{\Lambda}$ and $\mathcal{M}$ is actually physical motivated, we set

$$\pi_{\rm PE}(\mathcal{M}, q, \tilde{\Lambda}) = \pi(\mathcal{M}, q, \tilde{\Lambda}|\epsilon), \tag{C8}$$

which then implies that

$$p(d|\epsilon) = \int p(\mathcal{M}, q, \tilde{\Lambda}|d) \; d\mathcal{M} \; dq \; d\tilde{\Lambda}. \tag{C9}$$

Since the chirp mass is often very-well constrained by parameter estimation, we take the median value of chirp mass $\bar{\mathcal{M}}$, and rewrite our posterior distribution as

$$p(\mathcal{M}, q, \tilde{\Lambda}|d) = p(q, \tilde{\Lambda}) \; \delta(\mathcal{M} - \bar{\mathcal{M}}), \tag{C10}$$

which simplifies the integral to

$$p(d|\epsilon) = \int p(\bar{\mathcal{M}}, q, \tilde{\Lambda}|d) \; dq \; d\tilde{\Lambda}. \tag{C11}$$

We note that we can also express the tidal deformability as a function of $\bar{\mathcal{M}}$, q and ϵ which removes another dimension from our integration

$$p(d|\epsilon) = \int p(q, \tilde{\Lambda}(\bar{\mathcal{M}}, q, \epsilon)|d) \; dq, \tag{C12}$$

which is now just an integral over mass ratio q.

We evaluate the probability density of our marginalized posterior samples using a bounded kernel-density estimate, and parameterise our equation of state using the spectral parametersiation of Lindblom (2010). We use uniform priors for each equation of state parameter, and enforce a minimum TOV mass $M_{\rm TOV} \geq 2.0 M_{\odot}$ and causality.

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